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A foundation model for clinical-grade computational pathology and rare cancers detection

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Model name:

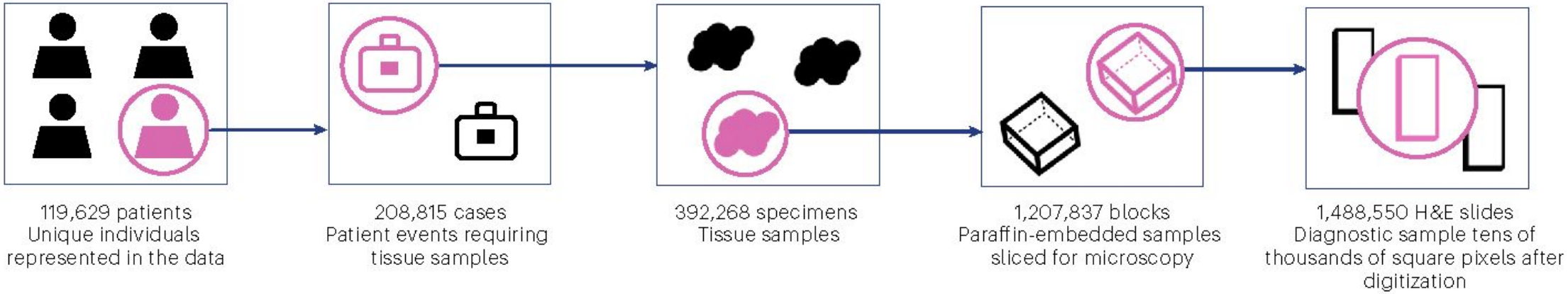
Virchow

Main Result/Insight

we demonstrate that a large foundation model enables pan-cancer detection, achieving 0.95 specimen-level area under the (receiver operating characteristic) curve across nine common and seven rare cancers. Furthermore, we show that with less training data, the pan-cancer detector built on Virchow can achieve similar performance to tissue-specific clinical-grade models in production and outperform them on some rare variants of cancer.

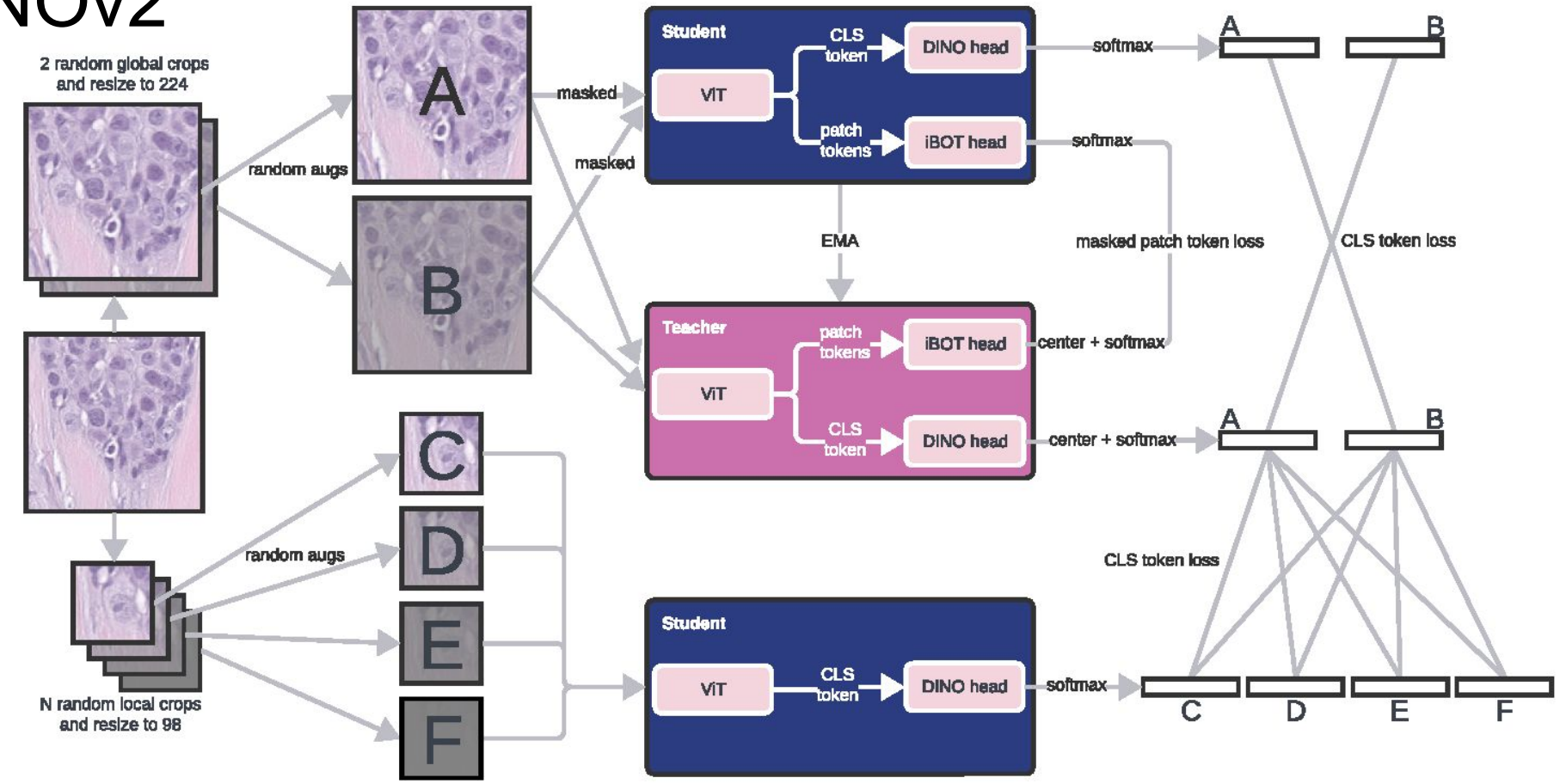
Cancer detection was found to scale approximately logarithmically with the number of model parameters (Fig. 2g, top); although performance scaled with the number of training tile samples, the trend (Fig. 2g, bottom) suggests diminishing returns.

Dataset



Virchow was trained on 2 billion tiles sampled randomly with replacement from approximately 13 billion available tissue tiles.

Pretraining: ViT-H trained with DINOv2



Embeddings: 2560

Evaluation 1: Pan-cancer detection

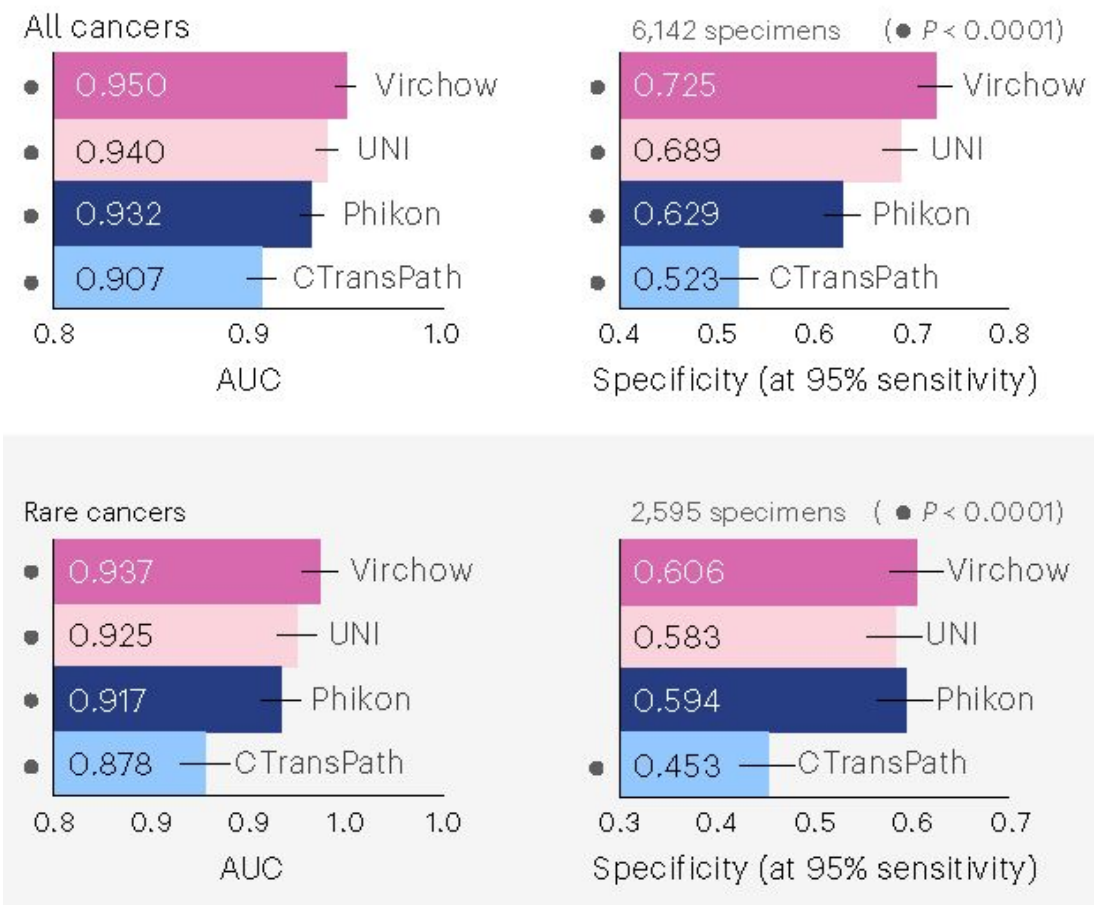
- Stratified performance for 9 common and 7 rare cancers

- Tile level embeddings to slide level embeddings using Agata aggregator
  Aggregator is a linear attention module

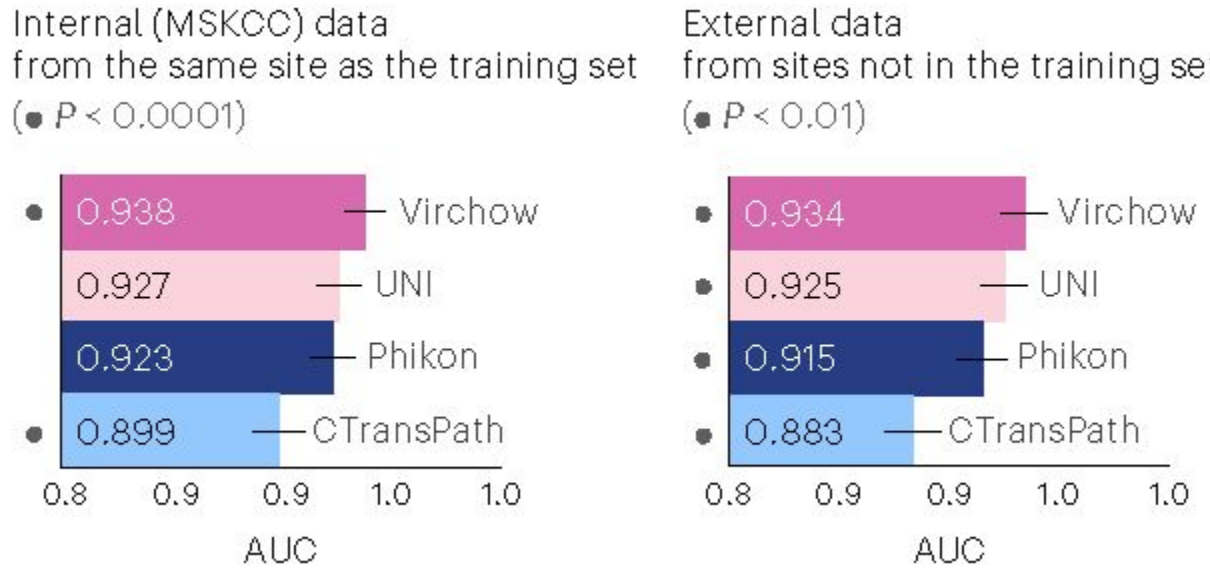
- MSKCC hold-out and other centers

- UNI, Phikon, CTransPath

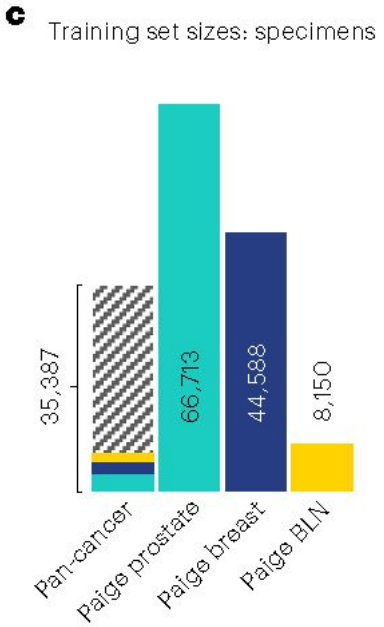
Across cancers

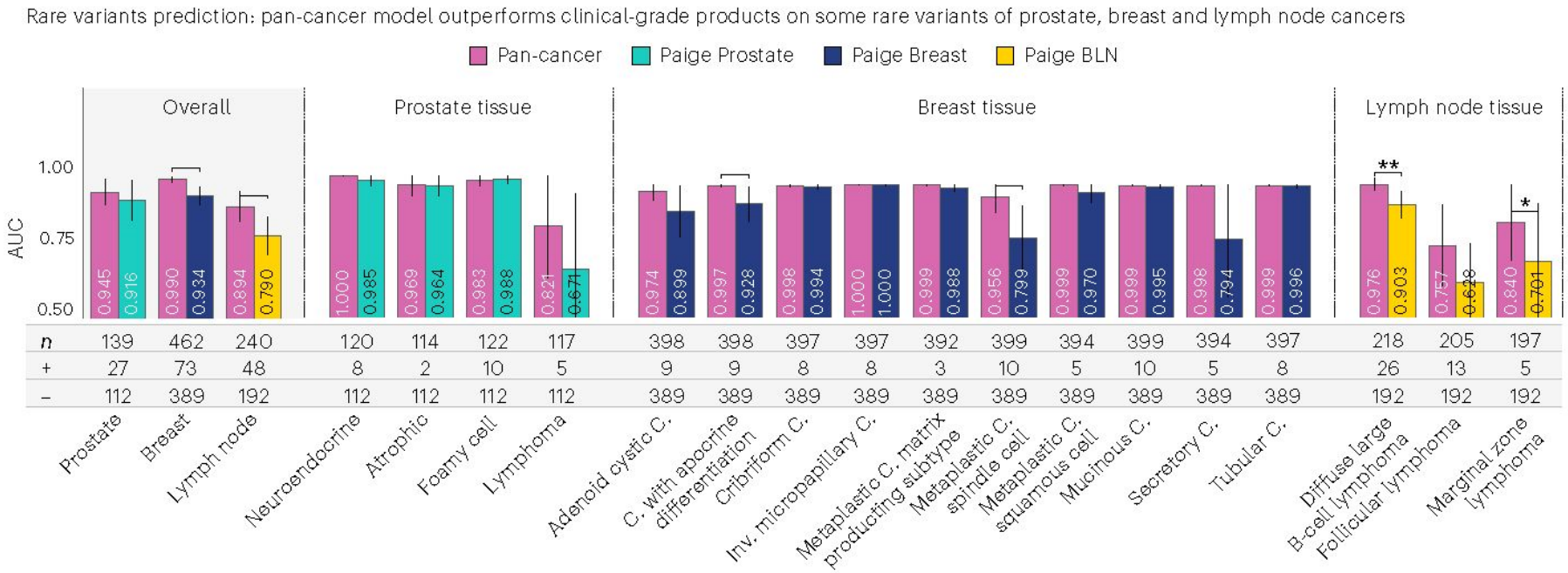


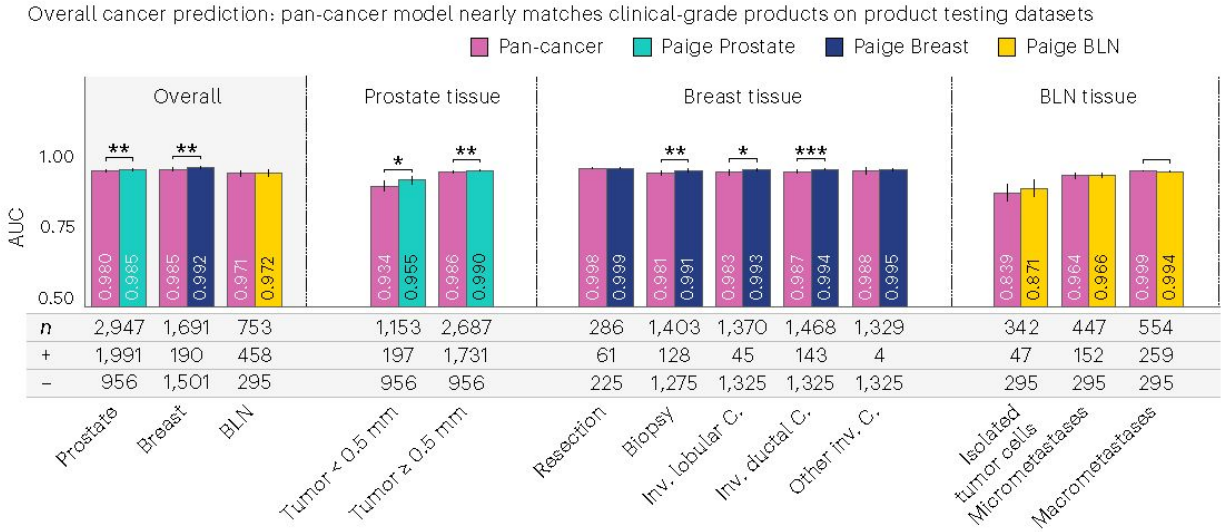
Across datasets



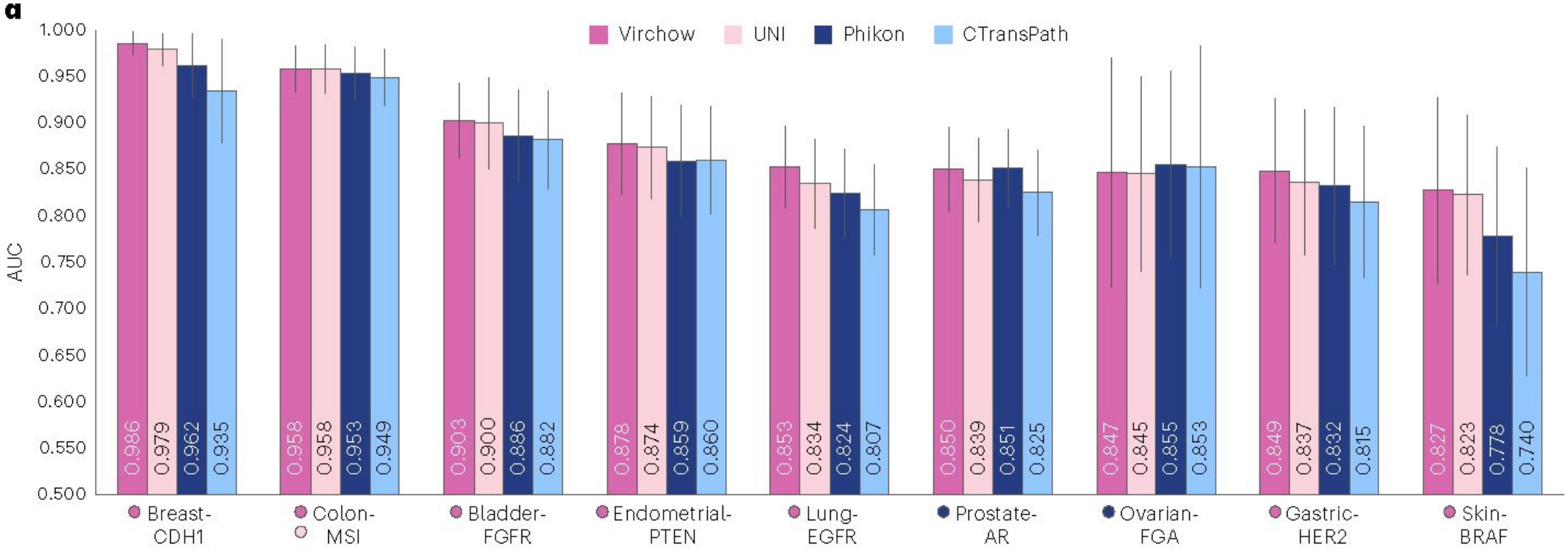
Evaluation 2: Comparison to commercial software



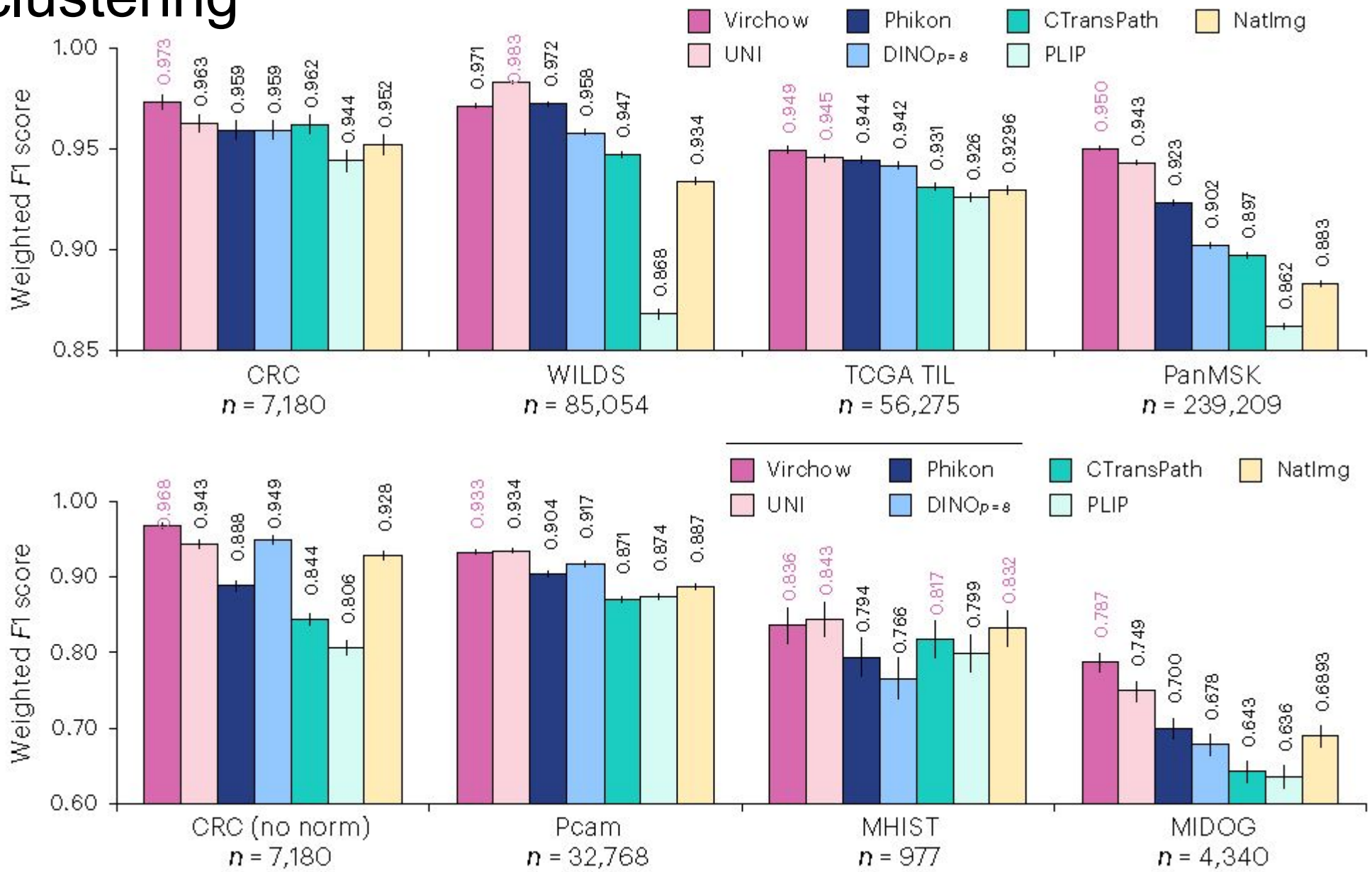




Evaluation 3: Biomarker prediction



Evaluation 4: Linear probing of tiles and clustering



Training data from 1 center – more centers = better embeddings?

Pathology specific training paradigms can lead to improvements?

Design space exploration was not done for aggregators.

Data size is defined by #WSIs or #tiles?